Looking under the streetlight of economic theory: opacity and visibility of transactions on the French electricity market during the European energy crisis

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Abstract

This communication examines transaction opacity in the French electricity market during the European energy crisis. It highlights the limits of dominant economic theories in informing effective mitigation policies, particularly with regard to the market's unexpected distributive effects. Knowledge framings, shaped by economic models, informed public debate and regulatory action while leaving over-the-counter transactions opaque. By analysing expert discourses, policy interventions, and structural market complexities, the paper shows how the visibility and opacity of economic exchanges shape political debate and state responses to crisis.

Keywords: electricity markets; opacity; ignorance; market devices; performativity; energy crisis. **Workshop themes:** the externalities of markets; the ignorance of markets.

In the context of what actors unanimously described as a "European energy crisis," these actors — and politicians in particular, given its considerable redistributive effects between producers and consumers — turned the sharp rise in spot prices and in the prices of forward contracts traded on wholesale markets into a public problem. On these markets, the formation of electricity prices became the object of extensive debate across political, industrial, media, and academic arenas, amid an unprecedented circulation of reasoning schemes drawn from the microeconomic theory of electricity markets — in particular the merit-order curve and marginalist accounts of price formation. In this setting, the auction mechanisms of the energy exchanges served actors as an argumentative and pedagogical support, being the sole publicly accessible window onto electricity exchange.

Yet during and after the crisis, despite the visibility of prices formed by auction algorithms, no observer was able to quantify the market's (re)distributive effects over the period. A large share of transactions — between producers, traders, and clients, on the wholesale market and at the level of supply contracts — is not made public. Under an idealised market regime, this absence of transparency poses no problem for the delivery of a socio-economic optimum: through continuous arbitrage, the truth of prices (that of real time, at the moment an electron is injected into or withdrawn from the grid) is, in theory, disclosed to market actors, the market figuring as a site of veridiction; at the horizon of a stabilised electricity system, actors' anticipations are perfectly elicited by markets rendered liquid through the proliferation of financial derivatives [1]. Price levels are held to converge progressively along a gradient running from the most decentralised to the most centralised, and from the longest to the shortest maturity [2]. In short, forward prices align with anticipated spot prices at delivery, and over-the-counter (decentralised) trades align with the centralised references of the exchanges. Yet, as the recent history of electricity markets attests, even supposing it real, this progressive convergence of price references is in no way a linear process. The superimposition of price references pointing to distinct temporalities renders the reading of displayed prices complex, particularly when, in disturbed conditions, past transactions crystallise anticipations rendered obsolete by recent developments.

In the crisis, and given the scale of price variation, the French government and its European counterparts opened reform discussions. Several solutions were considered, some even calling into question the marginalist principle of price formation. Ultimately, the measures adopted addressed only locally the risks of fragmenting the network of actors liable to destabilise the market paradigm that configures it: a tariff shield, regulated access to historic nuclear electricity (ARENH), and the taxation of profits to mitigate or even suppress the potential economic and political effects of prices. These interventions were nonetheless deployed under a certain blindness, owing to the lack of transparency over the vast majority of transactions — because of the preservation of commercial secrecy and, beyond it, the complexity of the architecture of contemporary electricity markets. Likewise, public debate over the causes and the modalities of mitigation was largely framed by the transactions made visible and by an economic theorisation that simplified the reality of exchange and its economic effects. The only precise knowledge available at the close of the crisis concerns public expenditure and state revenue, through the assessments of the Court of Auditors — but these capture only a small part of the economic effects.

We propose to treat electricity markets as sociotechnical assemblages, produced by a social engineering grounded in economic expertise [3]. Market exchange, and price formation, are equipped by auction algorithms and by equivalence devices that form prices out of other prices [4]. We draw on the questions raised by science and technology studies concerning the limits of expertise [5], whose advances enable the creation of innovative technologies without a comparable capacity to assess their effects or their reliability [6] — a perspective already mobilised to interrogate derivatives markets [7]. Electricity markets may be considered innovative technologies that demand a very high level of knowledge and modelling, and that call for a necessary fragmentation in their design [8]. This bottom-up, brick-by-brick construction — better, this serialisation of “micro-frames” [9] — opens interstitial spaces of arbitrage that escaped market designers and into which actors rush; for, unlike most hazardous technologies whose users generally seek to contain unpredictable effects, the exploration of these gaps offers further opportunities for profit [8]. The accretion of these micro-overflows produces an unpredictability of the markets’ macro-level economic effects. While the framing of transactions by centralised auction algorithms and the ex post control of market manipulation have improved considerably through the creation of calculative centres [10] — repositories of an expertise constructed to capture divergences between what these assemblages actually produce and what they are meant to perform — they bear only on part of the exchange, and ignore the over-the-counter trades that still constitute the vast majority of it. Crisis situations may also be considered extreme situations outside the field of established knowledge, where the validity assumptions of economic theory do not hold — the principle of a counter-performativity of theory [8]. Knowledge may further be limited by specific declensions of the technology, in particular through combinations with complementary techniques that increase the system’s complexity, as in the French market, where a substantial share of electricity exchange passes through a regulated mechanism, regulated access to historic nuclear electricity [11].

A first framing concerns the saturation of the politico-administrative and media space by the explanation of electricity price rises (spot and forward) through (multifactorial) tensions on gas markets: economic theory, in its most distilled and actor-accessible form, was widely mobilised in public debate to tie this rise to that of gas prices, leaving in shadow other causes such as the low availability of the French nuclear fleet or the exercise of market power by certain intermediaries. A second framing concerns the simplification that assimilates all electricity exchange to the visible exchanges on the market, passing over forward and over-the-counter trades in silence. This framing played an important role in the administration’s justification of the extension of the ARENH mechanism in March 2022, a decision strongly contested by EDF. It also informed the introduction of a tax on inframarginal rents and the initial estimates of its yield, out of all proportion to the revenue base eventually observed ex post. The opacity surrounding the economic practices of suppliers, intermediaries between wholesale and retail markets, likewise greatly hampered the monitoring of suppliers’ financial risks, as well as of their crisis profits. Unable to estimate the prices of supply contracts, the administration had to collect firms’ energy invoices in order to calculate aid. Finally, the opacity of transactions precludes any precise economic assessment of the crisis’s costs for firms, as of the economic balance for suppliers.

Within the workshop, our contribution articulates the theme of the externalities of markets with that of their ignorance. Precisely, it seeks to specify the practices, rules, and devices of visibilisation and opacification — the production of non-knowledge — of economic exchange on the French electricity markets, together with their simplified understanding within political debate. We assess the extent to which these "knowledge framings" are inherited from economic theories of the market and the implicit assumptions they carry; we account for how these practices translate into acts of regulation and political intervention in market functioning; and we interrogate the strategic character of the framings actors operate in the course of debate, since rendering economic exchange visible or invisible is itself a means of influencing public debate and public action to strategic ends. We draw on the analysis of a vast corpus of publications by sector actors and the specialised economic press, together with interviews conducted (and forthcoming) with members of these institutions.

Our argument may be summarised thus. If economic theory and electrical engineering have largely fed the design of market algorithms and shaped exchange, they also shape the way markets are known, understood, and controlled by public authorities. This theorisation functions like a streetlight: it depends on a sociotechnical network — algorithms, platforms, regulation, academic circles — for its construction, its operation, and its maintenance; but in return it offers only a reduced visibility, direct or indirect, confined to the beings that ground that network. By incorporating bilateral contracts little or not at all, and by stylising to the extreme the various maturities of forward markets, this theorisation cannot, by construction, render them visible. When dysfunctions become politicised, actors point now to regulatory failures, now to the publicly decried operation of a marginalist price mechanism. This opacity translates into a paralysis of the corrective action the state might undertake in crisis situations, where prices diverge significantly from prior anticipations, creating large differences between forward contracts according to the moment at which they were concluded.

References

[1] Jefferis, C. (2018). Financial performativity as evidence of immanence: The phenomenology of liquidity crisis in contemporary markets for risk. *Journal of Cultural Economy*, 11(4), 291–302.

[2] Léautier, T.-O. (2018). *Imperfect markets and imperfect regulation: An introduction to the microeconomics and political economy of power markets*. Cambridge, MA: MIT Press.

[3] Breslau, D. (2013). Designing a market-like entity: Economics in the politics of market formation. *Social Studies of Science*, 43(6), 829–851.

[4] Çalışkan, K. (2007). Price as a market device: Cotton trading in Izmir Mercantile Exchange. In M. Callon, Y. Millo, & F. Muniesa (Eds.), *Market devices* (pp. 241–260). Oxford: Blackwell.

[5] Bloor, D. (1991). *Knowledge and social imagery* (2nd ed.). Chicago: University of Chicago Press.

[6] Downer, J. (2017). The aviation paradox: Why we can 'know' jetliners but not reactors. *Minerva*, 55(2), 229–248.

[7] MacKenzie, D. (2011). The credit crisis as a problem in the sociology of knowledge. *American Journal of Sociology*, 116(6), 1778–1841.

[8] Rilinger, G. (2024). *Failure by design: The California energy crisis and the limits of market planning*. Chicago: University of Chicago Press.

[9] Callon, M. (1998). An essay on framing and overflowing: Economic externalities revisited by sociology. *The Sociological Review*, 46(1_suppl), 244–269.

[10] Beunza, D., & Garud, R. (2007). Calculators, lemmings or frame-makers? The intermediary role of securities analysts. *The Sociological Review*, 55(2_suppl), 13–39.

[11] Reverdy, T., & Breslau, D. (2019). Making an exception: Market design and the politics of re-regulation in the French electricity sector. *Economy and Society*, 48(2), 197–220.